

Lec. 02.

## Vector Properties

$\|\bar{u}\|$  = Norm of a vector

$\|\bar{u}\| = 1$   $\rightarrow$  unit norm vector

$\hat{u} = \frac{\bar{u}}{\|\bar{u}\|}$  = unit norm vector for any vector  $\bar{u}$

Ex: For signal vector  $\bar{u}$

$$\|\bar{u}\|^2 = \underbrace{u_1^2 + u_2^2 + \dots + u_n^2}_{\text{Energy of signal}}$$

$$\bar{u} = \begin{bmatrix} -1 \\ 3 \\ -2 \\ 1 \end{bmatrix}$$

$$\|\bar{u}\| = \sqrt{1 + 3^2 + (-2)^2 + 1} \\ = \sqrt{15}$$

$$\hat{u} = \frac{\bar{u}}{\|\bar{u}\|} = \frac{1}{\sqrt{15}} \begin{bmatrix} -1 \\ 3 \\ -2 \\ 1 \end{bmatrix}$$

unit norm vector

## Distance

$\bar{u}, \bar{v}$

$$\|\bar{u} - \bar{v}\|$$

Distance  
between  
vectors  $\bar{u}, \bar{v}$ .

$$= \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \dots + (u_n - v_n)^2}$$

$$= \sqrt{\langle \bar{u} - \bar{v}, \bar{u} - \bar{v} \rangle}$$

## Cauchy - Schwarz inequality

### CS Property

$$|\langle \bar{u}, \bar{v} \rangle| \leq \|\bar{u}\| \cdot \|\bar{v}\|$$

$$|\langle \bar{u}, \bar{v} \rangle|^2 \leq \|\bar{u}\|^2 \cdot \|\bar{v}\|^2$$

Real vectors :  $(\bar{u}^T \bar{v})^2 \leq \|\bar{u}\|^2 \cdot \|\bar{v}\|^2$

Complex vectors :

$$|\bar{u}^H \bar{v}|^2 \leq \|\bar{u}\|^2 \|\bar{v}\|^2$$

Simple Proof :

Consider  $\bar{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$   $\bar{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

Real vectors

$$\bar{u}^T \bar{v} = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = u_1 v_1 + u_2 v_2$$

$$\begin{aligned} (\bar{u}^T \bar{v})^2 &= (u_1 v_1 + u_2 v_2)^2 \\ &= u_1^2 v_1^2 + u_2^2 v_2^2 \\ &\quad + 2 u_1 u_2 v_1 v_2 \end{aligned}$$

Geometric mean  $\leq$  Arithmetic mean

Gm  $\leq$  Am

$$2 u_1 u_2 v_1 v_2 \leq u_1^2 v_2^2 + u_2^2 v_1^2$$

$$\begin{aligned} (\bar{u}^T \bar{v})^2 &\leq u_1^2 v_1^2 + u_2^2 v_2^2 \\ &\quad + u_1^2 v_2^2 + u_2^2 v_1^2 \\ &= (u_1^2 + u_2^2)(v_1^2 + v_2^2) \\ &= \|\bar{u}\|^2 \cdot \|\bar{v}\|^2 \end{aligned}$$

$$|\bar{u}^H \bar{v}|^2 \leq \|\bar{u}\|^2 \cdot \|\bar{v}\|^2$$

Example

$$\bar{u} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$\bar{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$(\bar{u}^T \bar{v})^2 = \begin{bmatrix} 1 & -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$= (1 - 2 + 3 - 4)^2$$

$$= 4$$

$$\|\bar{u}\|^2 = 4$$

$$\|\bar{v}\|^2 = 1^2 + 2^2 + 3^2 + 4^2 = 30$$

$$\therefore \|\bar{u}\|^2 \cdot \|\bar{v}\|^2 = 4 \times 30 = 120$$

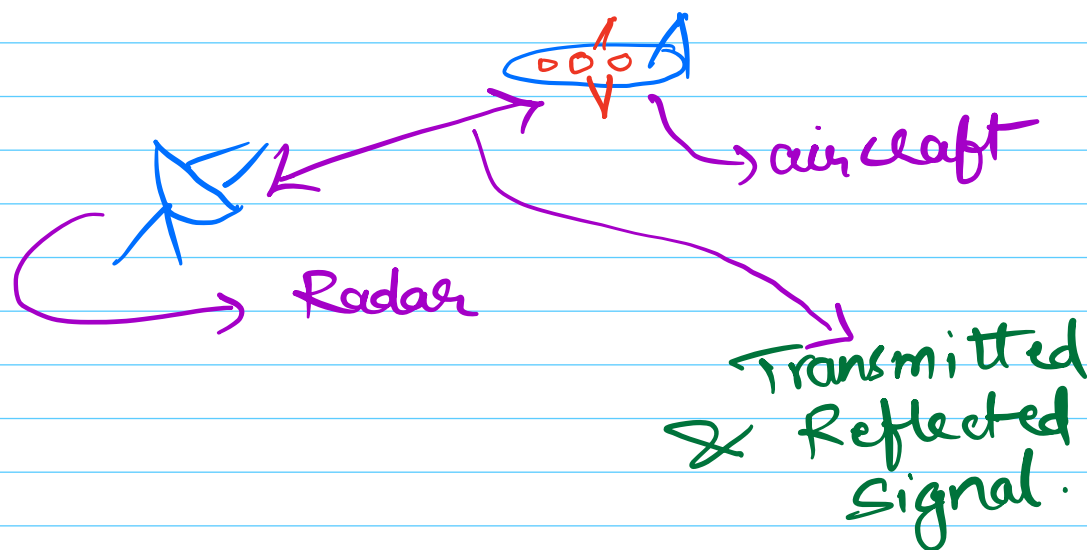
$$> 4$$

$$= |\bar{u}^T \bar{v}|^2$$

$\bar{u}^T \bar{v}$  = correlation between  
2 signals  $\bar{u}, \bar{v}$

This has many interesting  
applications.

RADAR correlation  
or inner product  
→ Radar Signal  
processing.



Transmit  
Signal vector  $\bar{x} = \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix}$  samples of  
Tx signal

$$\bar{y} = \begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(N-1) \end{bmatrix}$$

Received signal

Measured after delay

if object present :

$$\bar{y} = \bar{x} + \bar{w}$$

Rx signal

observed signal

Transmitted

Noise

AWGN

$\bar{y}$  will be highly correlated with  $\bar{x}$

$\Rightarrow$  inner product

$|\bar{x}^H \bar{y}|$  is very high.

if object is absent :

Target

$$\bar{y} = \bar{w}$$

Noise

similarity b/w  $\bar{y}$  &  $\bar{x}$  is low.

To determine presence / absence  
of target

→ Correlate received signal  
with transmitted signal.

$$|\bar{x}^H \bar{y}|^2 \geq \gamma \rightarrow \text{Threshold gamma}$$

$\Rightarrow$  Target is present

$$|\bar{x}^H \bar{y}|^2 < \gamma$$

$\Rightarrow$  Target is absent

RADAR — Detection problem  
Binary Hypothesis  
Testing problem.

Similarity between signals

Voice →  $\bar{x}_1$ ,  $\bar{x}_2$   
Images →

↓  
To check similarity

$$\|x_1^H x_2\|^2 \geq \lambda \Rightarrow \text{similar}$$

$$< \lambda \Rightarrow \text{dissimilar.}$$